# MAKING SOLUTIONS AND DILUTIONS

OBJECTIVES

- To understand the difference between stock and working solutions
- To appreciate the importance of dilutions and gain practical experience using dilution calculations
- To make a series of stock and working solutions for use in future labs

INTRODUCTION

Scientists across disciplines often quantitatively express solute concentration in various other terms (or units). Some of the more commonly used solute concentration units are molarity ($M$), percent by mass ($mass\ \%$), parts per million ($ppm$), and mole fraction ($\chi$). Molarity is, perhaps, the most used concentration unit, but you will certainly encounter many of these, and others, throughout your experience as a scientist.

Table 1: A summary of some common ways to express solute concentration in solution

| Concentration Unit | Symbol | Equation | Units |
|---|---|---|---|
| Molarity | $M$ | $\frac{\text{moles of solute}}{\text{Liters of total solution}}$ | $\frac{mol}{L}$ |
| Mass % | % | $\frac{\text{mass of solute}}{\text{total mass of solution}} \times 100$ | % |
| parts per million | ppm | $\frac{\text{mass of solute}}{\text{total mass of solution}} \times 10^6$ | ppm |
| Mole Fraction | $\chi$ | $\frac{\text{moles of solute}}{\text{mol solute} + \text{mol solvent}}$ | none |

NOTE: THE 7 REQUIRED PRE-LAB CALCULATIONS ARE FOUND ON PAGES 6 AND 7 OF THIS MANUAL!

SOLUTIONS AND DILUTIONS

In practical laboratory situations, we often need large amounts of a particular "working" solution that we might use daily. It can be impractical to make this solution over and over (and over), again. In addition, multiple working solutions at various concentrations can

be made from a single stock solution. Other times, exceedingly small amounts (think: *micro-*, *nano-,* and *pico-*) of reagent are called for in a chemical process. In this scenario, it is unrealistic to make such small amounts of solution OR perhaps such small masses can't effectively be measured on a balance. Stock solutions and dilutions can help navigate these situations.

Stock solutions are used to save time, conserve materials, reduce storage space, and increase the accuracy with which lower concentration solutions are prepared. When stock solutions are prepared, it is often known that the solute will be stable in solution and won't break down appreciably over time.

Typically, stock solutions are prepared at standardized concentrations and in larger volumes so that we can:

1. Use it repeatedly (without having to make it again and again)

2. Dilute it to achieve smaller final solute concentrations in more complex mixtures

3. Avoid having to inaccurately weigh such small amounts on a balance

As an example, in the "Measuring an Equilibrium Constant Lab" it calls for a $0.2\ mM\ SCN^-$ solution ($0.2\ mM = 0.0002\ M$). To make this solution would require you weigh $1.9\ mg$ (or $0.0019\ g$) of $KSCN_{(s)}$ on the balance in lab. Most balances cannot be relied upon to accurately measure masses to that precision, so we will make a **$20\ mM\ SCN^-$ stock solution** and dilute it 100-fold to obtain a **$0.2\ mM\ SCN^-$ working solution**. You will then use that $0.2\ mM\ SCN^-$ solution in a series of reactions that allow for equilibrium constant determination.

**NOTE**: We call these more diluted solutions our "working" solutions and they help us achieve a "working" concentration of some chemical reagent in a reaction or more complex solution (i.e., a solution containing multiple solutes). When you become more comfortable making dilutions, the advantages of using stock solutions become readily apparent.

For any solution being prepared, we need to calculate a few bits of information to prepare the solution correctly:

1. The total volume of solution we want/need (in L or mL)

2. The final concentration of the solute and in the desired unit we will express this concentration (M, ppm, %, mole fraction, etc.)

3. The mass (in g) of solute we will add and the molar mass (g/mol) of the solute

4. The identity and perhaps the density (depending on concentration unit) of the solvent

One of the most common mistakes in laboratory experiments is making solutions incorrectly.

In this lab, you will prepare a series of **stock** and **working solutions** that you will use in the following weeks.

You will make four stock solutions. Distilled water ($dH_2O$) is the solvent, although some will utilize a solution which contains 0.5 M nitric acid ($HNO_3$). **Please read carefully to make sure you use the correct solvent for each solution.** Use clean wash/squirt bottles of $dH_2O$ as a solvent source and refill them from the white handled taps at the sink as needed.

1. **Instructions on using volumetric glassware and graduated cylinders**
    a. Rinse your glassware/cylinder with distilled water a few times before using to be sure your glassware is free of contaminants. Because $dH_2O$ is your solvent, it is okay if some residual $dH_2O$ remains in your glassware.
    b. Be sure to re-rinse glassware thoroughly with $dH_2O$ before re-using.
2. **How to make a molar solution**

The general process for making a solution involves the following series of steps (concentrations in molarity):

a. Calculate grams of solute needed based on desired Molarity ($M$):

$$\left(\frac{\text{mol solute}}{\text{L solution}}\right) \times final\ volume\ (L) = moles\ of\ solute$$

$$moles\ of\ solute\ (mol) \times Molar\ mass\ of\ solute\ \left(\frac{g}{mol}\right) = grams\ of\ solute\ in\ solution$$

b. Measure that calculated mass (in g) of solute on a CLEAN balance.
c. Transfer your massed solute into a volumetric flask.
d. Add approximately $\frac{1}{2}$ of your final volume of solvent and *completely* dissolve the solute.
e. Bring up to the final volume with solvent (note meniscus line).




(a) An amount of solute is weighed out on an analytical balance and then (b) A portion of the solvent is added to the volumetric flask. (c) The mixture is swirled until *all* of the solute is dissolved. (d) Additional solvent is added up to the mark on the volumetric flask.

**NOTE:** If your measured mass of solute does not precisely match the calculated amount, you MAY have to calculate the actual molarity of your final solution.

f. Example calculation for mass needed of $CuSO_4 \cdot 5\ H_2O_{(s)}$ to make 100 mL of an aqueous 0.2 $M\ Cu^{2+}$ solution:

$$\left(\frac{\text{mol solute}}{\text{L solution}}\right) \times final\ volume\ (L) = moles\ of\ solute$$

$$0.2\ \frac{mol}{L} \times 0.1\ L = 0.02\ mol$$

Calculate molar mass of $CuSO_4 \cdot 5\ H_2O_{(s)}$, remember to account for the hydrate atoms!

$Cu: 1\ \times 63.546\frac{g}{mol} = 63.546\frac{g}{mol}$ $\quad$ $S: 1 \times 32.065\frac{g}{mol} = 32.065\frac{g}{mol}$

$O: 9 \times 15.99\frac{g}{mol} = 143.91\frac{g}{mol}$ $\quad$ $H: 10 \times 1.008\frac{g}{mol} = 10.08\frac{g}{mol}$

Molar mass = $63.546\frac{g}{mol} + 32.065\frac{g}{mol} + 143.91\frac{g}{mol} + 10.08\frac{g}{mol} = 249.60\frac{g}{mol}$

$$moles\ of\ solute\ (mol) \times Molar\ mass\ of\ solute\ \left(\frac{g}{mol}\right) = grams\ of\ solute\ in\ solution$$

$$0.02\ mol\ \times \frac{249.60\ g}{1\ mol} = 4.992\ g$$

**3. How to make a dilution:**

a. The general process for making dilutions is to use the simple expression, $M_1V_1 = M_2V_2$, if using molar concentrations of solutions. Or, more generally, you can use $C_1V_1 = C_2V_2$ if using some other concentration unit ($C$ stands for concentration).

b. It is important for you to understand WHAT variable in the above equation you are solving for ($M_1$, $V_1$, $M_2$ or $V_2$).

c. It is also important to understand the difference between the final volume of solution and the amount of solvent added to dilute.

d. Example calculation for 100 mL of an aqueous 5 $mM\ Fe^{3+}_{(aq)}$ solution:

$M_1V_1 = M_2V_2$, write down what you know:

$M_1 = 20\ mM\ Fe^{3+}_{(aq)}$ $V_1 = ?$ $M_2 = 5\ mM\ Fe^{3+}_{(aq)}$ $V_2 = 100\ mL$

Solve for volume you need from the stock solution:

$$V_1 = \frac{M_2V_2}{M_1} = \frac{(5\ mM)(100\ mL)}{(20\ mM)} = 25\ mL$$

You would mix 25 mL of $20\ mM\ Fe^{3+}_{(aq)}$ solution with 75 mL of nitric solution.

PROCEDURE

A. Making stock solutions

Make the following four stock solutions in the provided volumetric glassware using the general directions in the introduction. Briefly, you will weigh out each calculated solid amount of solute, transfer to the appropriately sized volumetric flask, add about half the total volume of the noted solvent below to dissolve, and then carefully fill up to the mark on the volumetric flask.

**Please do not put any scoopulas directly in the stock bottles containing the solid solutes.** First, dispense your solid into a clean weigh boat and label. Then, transfer the amount of solid that you need to weigh out. If you have extra solid, pass this along to the next group or dispose of in the waste.

0.1 M $Fe^{3+}_{(aq)}$ 20 mM $Fe^{3+}_{(aq)}$ 20 $mM\ SCN^{-}_{(aq)}$

**Note: All stock solutions should be colorless!** If you notice a red, orange or yellow coloration in your solution, you have cross-contaminated somewhere and will need to remake that stock solution.

Store these solutions in plastic Nalgene bottles (three 100 mL). **Label each appropriately with solute name, concentration, initials, and date. Show your instructor you've properly labeled these as you will need them in future labs.**

(**PRE-LAB Calculations**) Show all calculations for the mass needed to make each solution listed below in the data/calculations section of your notebook.

a. Make 100 mL of an aqueous 0.1 $M\ Fe^{3+}$ solution from $Fe(NO_3)_3 \cdot 9\ H_2O_{(s)}$
Solvent: nitric acid solution

b. Make 100 mL of an aqueous 20 $mM\ Fe^{3+}$ solution from $Fe(NO_3)_3 \cdot 9\ H_2O_{(s)}$
Solvent: nitric acid solution

c. Make 100 mL of an aqueous 20 $mM\ SCN^-$ solution from $KSCN_{(s)}$
Solvent: nitric acid solution

B. Making dilutions from stock solutions

1. **Cleaning pipettes for preparing solutions**
   a. Obtain a 250 mL beaker, label this as waste container.
   b. Never pipette from stock bottles. Use clean and dry beakers to pour solutions and solvent into and pipette from there (be careful to properly keep track here).
   c. Use a pipettor to fill your pipette one quarter full of distilled water.
   d. Remove the pipettor from the pipette while tilting the pipette on its side. DO NOT ALLOW THE LIQUID TO TOUCH THE PIPETTOR.
   e. Twirl the pipette so all inside surfaces are rinsed.
   f. Empty the pipette into your waste beaker.
   g. Repeat this process twice with distilled water.

2. **Prepare dilutions:**

You will make a series of dilutions from some of the stock solutions you made above in Part A. Use $M_1V_1 = M_2V_2$ to calculate and make these dilutions. Use volumetric pipettes and clean glass volumetric flasks of the appropriate size to prepare these dilutions. Use the pre-mixed **nitric acid solution** to dilute the stock solutions.

4 $mM\ Fe^{3+}_{(aq)}$ 6 $mM\ Fe^{3+}_{(aq)}$ 8 $mM\ Fe^{3+}_{(aq)}$ 0.2 $mM\ SCN^-_{(aq)}$

**Note: All working/diluted solutions should also be colorless!** If you notice a red, orange or yellow coloration in any of your solutions, you have cross-contaminated somewhere and will need to remake that diluted solution.

Store these solutions in the same manner as you stored your stock solutions. **Label each appropriately with solute name, concentration, initials, and date. Show your instructor you've properly labeled these as you will need them in future labs.**

(**PRE-LAB Calculations**) Show all calculations for the volume of stock solution needed to make each solution listed below in the data/calculations section of your notebook.

From your stock $\mathbf{20\ mM\ Fe^{3+}_{(aq)}}$ solution:

a. Make 50 mL of an aqueous 8 $mM\ Fe^{3+}_{(aq)}$ solution
   Solvent: nitric acid solution

b. Make 50 mL of an aqueous 6 $mM\ Fe^{3+}_{(aq)}$ solution
   Solvent: nitric acid solution

c. Make 50 mL of an aqueous 4 $mM\ Fe^{3+}_{(aq)}$ solution
   Solvent: nitric acid solution

From your stock $\mathbf{20\ mM\ SCN^-}$ solution:

d. Make 100 mL of an aqueous 0.2 $mM\ SCN^-_{(aq)}$ solution
   Solvent: nitric acid solution

C. Calculations and Tables for next week (to be done in lab):

In lab next week, you will making seven new solutions from your working solutions that you made this week. These seven new solutions will combine $Fe^{3+}$ and $SCN^-$ together to enable us to experimentallydetermine the equilibrium constant ($K_c$) associated with the following reaction:

$$Fe^{3+}_{(aq)} + SCN^-_{(aq)} \leftrightarrow FeSCN^{2+}_{(aq)}$$

Before we perform this experiment next week, it is important to know the initial concentration of each component of your new solutions.

a. First, determine $[\boldsymbol{SCN^-}]_{\boldsymbol{initial}}$ for each of the following solutions in Table 2 below and be sure to convert your new concentration into units of Molarity

b. An important assumption that we will make next week is that $[\boldsymbol{SCN^-}]_{\boldsymbol{initial}} = [\boldsymbol{FeSCN^{2+}}]_{\boldsymbol{equilibrium}}$, so you should have the same concentrations of these two things listed in Table 2:

Table 2: Calibration Solutions for the "Determining an equilibrium Constant, $K_c$" Lab

| **Solution #** | **Reagent volume: $0.11\ M\ Fe^{3+}$** | **Reagent Volume: $0.20\ mM\ SCN^-$** | **Reagent Volume: $0.5\ M\ HNO_3$** | **Final Volume** | $[SCN^-]_{in}$ $(in\ M)$ | $[FeSCN^{2+}]_{eq}$ $(in\ M)$ |
|---|---|---|---|---|---|---|
| **Blank** | 10.0 $mL$ | 0.0 $mL$ | 20.0 $mL$ | 30.0 $mL$ | | |
| **1** | 10.0 $mL$ | 5.0 $mL$ | 15.0 $mL$ | 30.0 $mL$ | | |
| **2** | 10.0 $mL$ | 10.0 $mL$ | 10.0 $mL$ | 30.0 $mL$ | | |
| **3** | 10.0 $mL$ | 20.0 $mL$ | 0.0 $mL$ | 30.0 $mL$ | | |

c. Determine $[Fe^{3+}]_{in}$ and $[SCN^-]_{in}$ for each of the following solutions that we will also make next week, and fill in the Table 3.

Table 3. Reaction Solutions for the "Determining an equilibrium Constant,$K_c$" Lab

| **Solution #** | **Reagent volume: $Fe^{3+}$** | **Reagent volume: $0.2\ mM\ SCN^-$** | **Final Volume** | $[Fe^{3+}]_{in}$ | $[SCN^-]_{in}$ |
|---|---|---|---|---|---|
| **4** | 10.0 $mL$ of 4.00 $mM$ | 10.0 $mL$ of 0.2 $mM$ | 20.0 $mL$ | | |
| **5** | 10.0 $mL$ of 6.00 $mM$ | 10.0 $mL$ of 0.2 $mM$ | 20.0 $mL$ | | |
| **6** | 10.0 $mL$ of 8.00 $mM$ | 10.0 $mL$ of 0.2 $mM$ | 20.0 $mL$ | | |

d. Next week, you will use the concentrations in the above tables to generate your calibration curve and generate ICE tables for $K_c$ determination.

e. Show all calculations and include these tables in the data/calculations section of your notebook.

BEFORE YOU LEAVE LAB

- Clean up your bench and common areas.
- Return all rinsed and clean glassware to the cart.
- Make sure all your solutions are properly labeled and stored in your lab drawer.
- No conclusion in your notebook is necessary for lab this week.

You should make sure you have an understanding for how you will make the 7 solutions for the equilibrium lab next week, and you should know the concentration of the components outlined in Tables 2 and 3 above.